

WORLD REALIZABILITY & EPISTEMIC HORIZONS

Manifesto and Research Programme

WREH-00 — version 0.1

Founding Research Programme

First technical target: **WR-I — Response-Defined Worlds**

Programme principle: *Responses constrain worlds; they do not automatically identify one.*

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1. Why this programme

Modern physics is extraordinarily successful at predicting observable phenomena. Modern mathematics provides languages, structures, and methods capable of describing the world with exceptional precision.

Yet the success of a mathematical model does not by itself establish that experiment has identified a unique ontology underlying that model.

The same finite body of observations may be compatible with several distinct microscopic structures, global histories, geometries, topologies, or other realizations.

The fundamental question can therefore be posed differently. Not only, “What is the world really like?”, but:

Which worlds remain compatible with everything physically accessible to observation?

and then:

What can shrink this class, make it rigid, or isolate a unique realization?

The **World Realizability & Epistemic Horizons (WREH)** programme is founded to study these questions mathematically and physically.

2. The primary object

Let $\mathcal{W}$ be a class of admissible global realizations, provisionally called *worlds*. Let $\mathcal{P}$ be a class of physically admissible observational or experimental protocols.

Each protocol $P \in \mathcal{P}$ defines a response

$$R_P : \mathcal{W} \rightarrow \mathcal{Y}_P.$$

Two worlds are indistinguishable relative to the accessible protocol class when

$$W_1 \sim_{\mathcal{P}} W_2 \iff R_P(W_1) = R_P(W_2) \quad \forall P \in \mathcal{P}.$$

Physical data D determine a set of compatible realizations

$$\mathfrak{W}(D) = \{W \in \mathcal{W} : W \text{ is compatible with } D\}.$$

The central object of WREH is not necessarily an individual W , but the structure

$$\mathfrak{W}(D).$$

We ask about its size, components, fibres, boundaries, rigidity, connectivity, topology, stability, and its evolution under new observations.

3. A world may be specified by a class rather than an object

If

$$|\mathfrak{W}(D)| > 1,$$

the available data do not imply a unique global realization.

This may mean several very different things:

- a unique world exists, but the data are presently insufficient;
- several realizations are indistinguishable under all physically admissible observations;
- the theory itself specifies only a realization class;
- a particular realization is fixed by a new physical event;
- no global completion exists at all, despite consistency of every finite observational subset.

WREH does not choose among these interpretations in advance. Its task is to build a language in which they can be mathematically distinguished.

4. Refinement of knowledge

New data generate a nested sequence

$$\mathfrak{W}(D_1) \supseteq \mathfrak{W}(D_1, D_2) \supseteq \mathfrak{W}(D_1, D_2, D_3) \supseteq \dots$$

Unique identification

$$\bigcap_n \mathfrak{W}(D_1, \dots, D_n) = \{W^*\}.$$

A single admissible global realization remains.

Irreducible multiplicity

$$\left| \bigcap_n \mathfrak{W}(D_1, \dots, D_n) \right| > 1.$$

Even an arbitrarily rich class of accessible observations fails to isolate one world.

Loss of global realizability

It may be possible that

$$\forall n \quad \mathfrak{W}(D_1, \dots, D_n) \neq \emptyset,$$

yet

$$\bigcap_{n=1}^{\infty} \mathfrak{W}(D_1, \dots, D_n) = \emptyset.$$

Every finite set of requirements admits some world, but no single global world satisfies all requirements simultaneously. This possibility makes **global realizability** a mathematical problem in its own right.

5. The observable past and a unique past are not the same thing

The present world contains an enormous body of records of the past:

$$D_{\text{past}} = \{\text{memories, archives, geological structures, photons, CMB, causal traces, ...}\}.$$

All admissible realizations must reproduce these records. This does not necessarily imply uniqueness of the hidden history.

There may be $W_1 \neq W_2$ such that

$$R_{\text{past}}(W_1) = R_{\text{past}}(W_2).$$

Hence

$$\boxed{\text{observable past} \neq \text{unique past ontology.}}$$

A new experiment need not “change the past.” It may only remove some previously admissible global histories:

$$\mathfrak{W}_{\text{old}} \longrightarrow \mathfrak{W}_{\text{new}} \subseteq \mathfrak{W}_{\text{old}}.$$

6. Four realization scenarios

If a new experiment produces an outcome r ,

$$\mathfrak{W}(D) \longrightarrow \mathfrak{W}(D, r),$$

WREH distinguishes at least four conceptually different scenarios.

BRANCH

Admissible realizations physically branch:

$$[W] \rightarrow \{W_1, W_2, \dots\}.$$

CONTEXTUALIZE

The physically specified object belongs to a particular experimental context:

$$(P, r) \rightarrow X_{P,r}.$$

A unique global underlying object need not be required.

SELECT

After the outcome, only one global realization remains:

$$\mathfrak{W}(D, r) = \{W^*\}.$$

REMAIN A CLASS

Even after refinement,

$$|\mathfrak{W}(D, r)| > 1,$$

and the realization class is the maximally specified physical object available to the observer.

WREH does not assume in advance which of these scenarios is realized by nature.

7. World boundaries as boundaries of realization space

The phrase “boundary of the Universe” usually evokes a spatial wall. WREH studies a more general object:

$$\partial\mathfrak{W}.$$

The relevant boundary may live not in coordinate space, but in the space of admissible worlds. We provisionally distinguish

$$\partial_{\text{causal}}, \quad \partial_{\text{observational}}, \quad \partial_{\text{thermodynamic}}, \quad \partial_{\text{gravitational}}, \quad \partial_{\text{model}},$$

and the stronger object

$$\boxed{\partial_{\text{realizability}}},$$

beyond which no admissible global realization of the stated class exists.

The speed of light, entropy, quantum restrictions, gravity, horizons, and conservation laws may be treated as different constraints on the shape of admissible world space. WREH does not assume that currently known physical restrictions are already fundamental walls of reality itself.

8. Epistemic horizon

An observer is part of the world the observer is trying to know. Consequently, the class of admissible experiments need not distinguish all mathematically distinct realizations.

We use **epistemic horizon** for a situation in which

$$W_1 \neq W_2,$$

but

$$R_P(W_1) = R_P(W_2) \quad \forall P \in \mathcal{P}_{\text{admissible}}.$$

The difference exists at the level of the adopted mathematical model, yet no admissible internal experiment can turn that difference into a distinguishable response.

The term *epistemic horizon* already appears in contemporary literature and is not claimed as a new term by WREH. The programme-specific object is its connection to **global world realizability**, response-defined equivalence classes, and the geometry of admissible world space.

9. The programme's central question

In its limiting form, the programme asks:

Are some differences between possible worlds physically indistinguishable in principle?

and, more strongly:

Is there a genuine boundary to the set of physically realizable worlds?

This is not a question about where the edge of space is located. It is a question about **which global structures can exist at all under stated physical constraints and observed responses.**

10. Planned paper sequence

Code	Working title	Primary task
WREH-00	World Realizability & Epistemic Horizons — Manifesto and Research Programme	Founding document: objects, terminology, paper map, and claim firewalls.
WR-I	Response-Defined Worlds: Finite-Resolution Equivalence of Global Realizations	Formalize $(\mathcal{W}, \mathcal{P}, R)$, world fibres, and the observational quotient.
WR-II	Finite Consistency and Global World Realizability	Separate finite consistency from existence of a global completion; study compactness and limits.
WR-III	Geometry of Admissible World Fibres: Refinement, Rigidity and Realizability Walls	Study the geometry of $\mathfrak{W}(D)$, refinement, rigidity, singular sets, and realizability walls.
WR-IV	Response-Conditioned Global Completion and the Status of the Past	Formalize completion of histories and separate construction order from internal time.
WR-V	Experiment and Realization Selection: Branch, Contextualize, Select, or Remain a Class	Formalize four ways a world class may change after a new outcome.
WR-VI	The Aquarium Bounds: Physical Constraints as Boundaries of World Space	Classify causal, thermodynamic, quantum, gravitational, and model walls.
WR-VII	Energy-Time Frontiers of World Realizability	Distinguish a realizability frontier from lifetime, cutoff, Planck scale, speed limits, and vacuum decay.
WR-VIII	The Cosmological Fibre: Universes Compatible with Our Universe Today	Formulate the problem for real CMB/BAO/SN/lensing responses and global cosmological realizations.
WR-IX	Epistemic Horizons: What Can an Observer Inside a World Ever Distinguish?	Distinguish unknown, currently unobservable, inaccessible in principle, and physically indistinguishable.

WREH-00 — Manifesto and Research Programme

The present programme document. It contains no new physical result. Its purpose is to define the research object, terminology, work sequence, and claim firewalls.

WR-I — Response-Defined Worlds: Finite-Resolution Equivalence of Global Realizations

The foundational mathematical paper. Primary objects:

$$(\mathcal{W}, \mathcal{P}, R), \quad \mathcal{W}/\sim_{\mathcal{P}}, \quad \mathfrak{W}(D).$$

The aim is to formulate the minimal mathematical structure needed to discuss non-uniqueness of a world relative to a finite class of observations.

WR-II — Finite Consistency and Global World Realizability

Study the distinction between finite compatibility and global existence. Central question:

$$\forall n \, \mathfrak{W}(D_1, \dots, D_n) \neq \emptyset \implies? \bigcap_n \mathfrak{W}(D_1, \dots, D_n) \neq \emptyset.$$

Expected tools include compactness, inverse/projective limits, extension problems, and explicit counterexamples.

WR-III — Geometry of Admissible World Fibres: Refinement, Rigidity and Realizability Walls

Study the geometry of $\mathfrak{W}(D)$: components, fibres, singular sets, refinement, rigidity, stability, and realizability walls. This is a natural mathematical contact point with Boundary Compensation, but no BC wall is automatically interpreted as a spacetime or physical-world boundary.

WR-IV — Response-Conditioned Global Completion and the Status of the Past

Study maps of the form

$$R \longrightarrow \mathfrak{W}(R).$$

Construction order is separated from internal physical time. The aim is to determine when a locally accessible response admits a global history, how unique it is, and what it can mean mathematically for a completion to contain a past preceding its reconstruction. The paper does not claim retrocausality or physical “creation of the past.”

WR-V — Experiment and Realization Selection: Branch, Contextualize, Select, or Remain a Class

Formalize the four scenarios

$$\boxed{BRANCH/CONTEXTUALIZE/SELECT/REMAIN\ A\ CLASS}.$$

Study the action of a new outcome $\mathfrak{W}(D) \rightarrow \mathfrak{W}(D, r)$ and conditions under which an experiment leads to branching, contextual realization, unique completion, or persistent non-identifiability.

WR-VI — The Aquarium Bounds: Physical Constraints as Boundaries of World Space

Treat physical restrictions as candidate walls of realization space:

$$c, \quad \hbar, \quad G, \quad S, \quad \text{causality,} \quad \text{conservation laws,} \dots$$

The central question is whether an observed boundary is a wall of physical reality or merely a wall of a particular description. The title *Aquarium Bounds* preserves the originating intuition, while the paper assumes neither an external space nor an observer outside the Universe.

WR-VII — Energy-Time Frontiers of World Realizability

Introduce candidate characteristics such as

$$E_* = \sup\{E : \mathfrak{W}_E \neq \emptyset\}, \quad T_* = \sup\{T : \mathfrak{W}_T \neq \emptyset\}.$$

The goal is to distinguish object lifetime, quantum speed limits, effective-theory cut-off, Planck scale, gravitational collapse, and vacuum lifetime from the stronger notion of a boundary of global realizability. No known energy scale is identified with E_* in advance.

WR-VIII — The Cosmological Fibre: Universes Compatible with Our Universe Today

Move to real cosmological responses:

$$R_{\text{CMB}}, \quad R_{\text{BAO}}, \quad R_{\text{SN}}, \quad R_{\text{lensing}}, \quad R_{\text{structure}}, \dots$$

The task is not to reconstruct one Universe, but the space of global realizations still compatible with current observations. Model parameters, global geometry, topology, causal accessibility, and fundamental underdetermination must be kept distinct.

WR-IX — Epistemic Horizons: What Can an Observer Inside a World Ever Distinguish?

The final paper of the first cycle. It studies the distinction between unknown, currently unobservable, inaccessible in principle, and not physically distinguishable. Central question: is there something “beyond the horizon,” or does the distinction itself lack operational meaning for an observer inside the world?

11. Future synthesis

After the technical core is completed, the programme anticipates a synthesis paper:

From Objects to Worlds: Limits of Mathematical and Physical Reconstruction

It should connect WREH results with inverse problems, quantum foundations, quantum field theory, general relativity, thermodynamics, cosmology, analogue and emergent gravity, hydrodynamic models, embedded observers, scientific realism, and limits of mathematical reconstruction. This synthesis should not precede the technical core.

12. Relation to Boundary Compensation

WREH grows out of questions that arose naturally in Boundary Compensation, especially response as primary object, finite-resolution equivalence, inverse fibres, refinement, rigidity, admissibility, walls, and identifiability.

However, WREH is **not BC-XXIII**.

Boundary Compensation studied controlled finite-dimensional response structures. WREH changes the primary object:

realization of response $\longrightarrow$ global realization of a world.

Therefore the transition requires a separate mathematically explicit bridge. No finite-dimensional BC structure automatically becomes spacetime, a physical world, energy, time, or a cosmological boundary.

13. Claim firewalls

The programme deliberately constrains its own claims. WREH does not assume in advance that:

- underlying reality does not exist;
- observation creates a physical world;
- experiment rewrites the past;
- the Universe literally branches;
- the many-worlds interpretation is correct;
- collapse is fundamental;
- time is emergent;

- a minimum length or minimum time exists;
- the Planck scale is a limit of world existence;
- the world has an external wall;
- nothing exists beyond the cosmological horizon;
- mathematical consistency is equivalent to physical existence;
- every mathematically admissible world is physically realizable;
- finite-resolution equivalence proves ontological equivalence.

Every such claim requires a separate physical bridge and independent proof or experimental support.

14. The WREH community

World Realizability & Epistemic Horizons Community

The community is founded as an open research forum for mathematicians, physicists, engineers, philosophers of science, and researchers in AI-assisted science who are interested in rigorous limits of physical reconstruction.

Its core principle is:

Do not conflate the observed, the mathematically admissible, and the ontologically asserted.

Research areas

- **Mathematical Foundations:** equivalence relations, quotient spaces, compactness, inverse problems, extension theory, inverse/projective limits, topology, and geometry of realization spaces.
- **Physical Realizability:** causality, entropy, gravitation, quantum constraints, stability, horizons, and physical admissibility.
- **Cosmological Reconstruction:** global geometry, topology, causal horizons, and observationally indistinguishable cosmologies.
- **Quantum and Measurement Foundations:** contextuality, state reconstruction, outcome selection, superselection, and operational equivalence.
- **Embedded Observers:** physical limits on observers that are themselves contained in the world under investigation.
- **Computational Experiments:** finite models, certified simulations, and reproducible counterexamples.
- **Critical Audit:** prior art, hidden assumptions, duplicate concepts, no-go results, and counterexamples.

15. Community rules

WREH welcomes strong hypotheses but requires their status to be explicit. Claims should, where possible, be marked as one of:

THEOREM, *MODEL*, *NUMERICAL EVIDENCE*, *PHYSICAL HYPOTHESIS*, *INTUITION*

No category should silently transform into another. Negative results are first-class results of the programme. A counterexample is more valuable than an elegant but false general hypothesis.

16. What counts as a contribution

The community is not restricted to positive theories. A substantive contribution may be:

- a theorem;
- a no-go theorem;
- a counterexample;
- an explicit model;
- a classification result;
- a uniqueness theorem;
- a non-identifiability theorem;
- a compactness or non-compactness result;
- an experimental proposal;
- a numerical certificate;
- a prior-art audit;
- a critique exposing a hidden assumption;
- a proof that a proposed WREH construction reduces to already known mathematics.

The last outcome also counts as success.

17. Relation to existing science

WREH is not intended as an alternative to modern physics. It treats the Standard Model, quantum field theory, general relativity, thermodynamics, cosmology, and other successful frameworks as sources of constraints on the space of admissible realizations.

The programme also recognizes substantial neighbouring fields: inverse problems, observational equivalence, scientific underdetermination, epistemic horizons, embedded-observer models, quantum contextuality, cosmological horizons, spacetime extendibility, analogue gravity, and emergence.

Novelty must therefore be established **for specific mathematical objects and theorems**, not for broad philosophical statements.

18. Scientific position

WREH starts from a minimal principle:

A physical response is evidence, but not automatically proof of a unique underlying realization.

If an experiment identifies a unique realization, that must be shown. If a class remains, the class must be studied. If no physical protocol can ever divide the class, that too must be demonstrated.

If a global realization is absent, one must distinguish

failure of model from failure of world realizability.

19. Research ideal

Instead of asking, “Which world do we want to call fundamental?”, the programme asks:

What exactly do accessible observations force us to assert?

and then:

What remains free after all of those constraints are imposed?

The remaining freedom may collapse to one world. It may never do so. The equivalence class may turn out to be the only physically meaningful entity. Genuine realizability walls may emerge. Each of these outcomes is scientifically informative.

20. Open invitation

World Realizability & Epistemic Horizons is not founded to defend a preselected picture of the Universe.

The community is invited to search for proofs, counterexamples, prior analogues, constraints, experimental discriminators, domains of principled non-identifiability, conditions for existence and uniqueness of global completion, and conditions under which the very question of a unique underlying world ceases to be physically decidable.

Our subject is not a catalogue of fantastical worlds.

Our subject is

the structure of the set of worlds that physics has not yet been able to distinguish.

And its boundary is

the place where admissible realizations end, not where our imagination ends.

WREH-00 — Founding Research Programme. First technical target: **WR-I — Response-Defined Worlds.**